

Office Attendance System

Administrator Operations Manual

Complete setup, enrollment, day-to-day operations, and troubleshooting reference for school administrators.

DEVICE MANUFACTURER

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Getting Started

What to Expect When the Device Arrives

When you receive your attendance device, it will be fully assembled and pre-configured with your school's details. The only things you need to do before it is ready for daily use are: connect it to your school WiFi network, and enrol fingerprints for your staff members. Everything else has already been done for you.

◆ NOTE

The device has no battery. It must be plugged into a power source whenever it is in use. Keep it plugged in at all times during the school day.

What Has Already Been Done For You

- The backend system has been deployed and is fully operational.
- Your school spreadsheet has been created and linked to the system.
- The device firmware has been flashed and is ready to use.
- The device has been pre-configured with your school ID.

Your Responsibilities as Administrator

- **First-time setup:** Connect the device to WiFi and enrol the master finger.
- **Staff management:** Enrol new staff, remove leavers, and replace fingerprints when needed.
- **WiFi maintenance:** Reconfigure the WiFi connection if the school network changes.
- **Record keeping:** Monitor and correct attendance records in Google Sheets as needed.
- **Monthly reports:** Review the automated monthly summary email sent to your inbox.

First-Time WiFi Setup

Powering On the Device

Plug the device into a wall socket or USB power source using the cable provided. Once powered on, the fingerprint sensor LED will begin glowing — this confirms the device is running. On first boot the device has no WiFi credentials saved and will enter setup mode automatically. The sensor LED will pulse purple to indicate it is waiting to be configured.

Connecting the Device to School WiFi

1. On your phone or laptop, open your WiFi settings and look for a network named **Attendance-Setup**.
2. Connect to **Attendance-Setup**. The password is **setup1234**.
3. Once connected, open a web browser and go to: **http://192.168.4.1**. A setup page will appear.
4. Enter your school WiFi name (SSID) and password into the form.
5. Click **Save & Connect**. The device will save the credentials and restart automatically.
6. Reconnect your phone or laptop to your normal school WiFi after this step.

◆ NOTE

If the setup page does not appear automatically after connecting to Attendance-Setup, type <http://192.168.4.1> manually into your browser's address bar.

■ WARNING

Do not share the Attendance-Setup password (setup1234) with staff. Anyone who knows it and has physical access to the device could reconfigure it.

Verifying the Connection

After the device restarts, check the sensor LED:

LED State	Meaning
Steady blue	Connected to WiFi successfully. Ready to use.
Steady purple	WiFi connection failed. Check your password and try again from Step 1.

✓ TIP

If the LED stays purple after restarting, the most common cause is a wrong WiFi password. Repeat Section 2.2 and re-enter the credentials carefully.

Enrolling the Master Finger

What the Master Finger Does

The master finger is a special administrator fingerprint enrolled on the device. Its primary purpose is to trigger the WiFi setup mode — useful when you need to change the school WiFi network. When the device recognises the master finger, it flashes yellow five times and launches the WiFi setup portal.

◆ NOTE

The master finger does not record attendance. Scanning it will never create an entry in the Attendance sheet. It is purely an admin tool. Enrol it using your own fingerprint, or the fingerprint of whoever is responsible for maintaining the device.

How to Enroll the Master Finger

Enrollment is managed from the Enrollment tab in the school spreadsheet. Make sure the device is powered on and its LED is steady blue before proceeding.

1. Open your school spreadsheet in Google Sheets.
2. Click on the tab labelled **Enrollment** at the bottom of the sheet.
3. Click on the first empty row below the header row.
4. Fill in the columns as shown in the table below.
5. Within about 10 seconds, the device LED will turn solid yellow. This means it has picked up the enrollment request and is waiting for a finger.
6. Place your finger firmly on the sensor and hold it still until the LED changes.
7. Lift your finger when the LED goes off briefly, then place the same finger again for the second scan.
8. If successful, the LED flashes green. The Status column will update to **registered** within a few seconds.

Field	Description
FingerID	Leave blank. The device fills this in after enrollment.
UniqueID	Enter any short identifier, e.g. MASTER-01.
Name	Enter a label, e.g. Admin Master.
Role	Type exactly: master
Command	Type exactly: register
Status	Leave blank.
Note	Leave blank.
CreatedAt	Optional — you may enter today's date for your records.

■ WARNING

Do not enrol the master finger with Role set to staff. If you do, it will record attendance instead of launching the WiFi portal.

Verifying the Enrollment

After enrollment, confirm the following in the Enrollment sheet:

- The FingerID column now contains a number (e.g. 1).
- The Status column reads **registered**.
- The Role column reads **master**.

To confirm it works, place the master finger on the sensor. The LED should flash yellow five times, then turn purple — the WiFi portal is now running. Since you do not need to reconfigure the WiFi right now, simply unplug and replug the device to restart it normally. The blue LED will return.

✓ TIP

Test the master finger immediately after enrollment so you know it works before you ever need it in an emergency.

Enrolling Staff Members

How Enrollment Works

All enrollment is managed from the Enrollment tab in your school's Google Sheet. You add a row with the staff member's details and a **register** command, and the device detects it automatically within about 10 seconds. It then guides the staff member through two finger placements, stores the fingerprint, and updates the sheet with the result.

✓ TIP

Have the staff member standing next to the device before you add the row to the sheet, so they are ready when the yellow LED appears.

Registering a New Staff Member

1. Open the school spreadsheet in Google Sheets and click the **Enrollment** tab.
2. Click on the first empty row below the last entry.
3. Fill in the columns as shown in the table below.
4. Within about 10 seconds, the device LED will turn solid yellow, indicating it is ready for the first scan.
5. Ask the staff member to place their finger firmly and flatly on the sensor and hold it still.
6. The LED will go off briefly. Ask the staff member to lift their finger and wait for the LED to turn yellow again.
7. Ask the staff member to place the same finger on the sensor again for the second scan.
8. If successful, the LED flashes green and the Status column updates to **registered** within a few seconds.

Field	Description
FingerID	Leave blank. The device fills this in automatically.
UniqueID	The staff member's ID (e.g. EMP-042). Must be unique per person.
Name	The staff member's full name (e.g. Kwame Mensah).
Role	Type exactly: staff
Command	Type exactly: register
Status	Leave blank.
Note	Leave blank.
CreatedAt	Optional — you may enter today's date for your records.

■ WARNING

Each staff member must use the same finger every time they scan — both for enrollment and for all future attendance scans. Choose a finger with a clear, undamaged print. Index fingers usually work best.

What Happens on the Device During Enrollment

LED State	Meaning	Action
Solid yellow (first)	Waiting for first finger placement.	Place the staff member's finger on the sensor.
Solid yellow (second)	First scan done — waiting for same finger again.	Ask the staff member to place the same finger again.
Green flash	Enrollment successful.	Confirm Status = registered in the sheet.
Red flash	Enrollment failed.	Check the Note column for an error code. Retry (see Chapter 7).

Verifying a Successful Enrollment

Confirm the following before the staff member walks away:

- The FingerID column contains a number.
- The Status column reads **registered**.

Ask the staff member to do a test scan immediately. Their Time-In should appear as a new row in the Attendance sheet within a few seconds. If no row appears, see Chapter 7.

Day-to-Day Operations

How Staff Use the Device

Staff use is simple: place their registered finger on the sensor when arriving at school, and place the same finger when leaving. Nothing else is required.

- A **green flash** means the scan was accepted.
- A **red flash** means the fingerprint was not recognised. The staff member should try again.
- The **first scan** of the day is recorded as Time-In.
- Every scan after that updates Time-Out. The last scan of the day is the final departure time.
- Once a scan is accepted, the staff member must wait at least **60 seconds** before scanning again. Scans within 60 seconds of the previous accepted scan are silently ignored.

Monitoring Attendance in Real Time

Open the school spreadsheet in Google Sheets at any time to see live attendance. New rows appear in the Attendance sheet within a few seconds of each scan. You do not need to refresh the page.

✓ TIP

Bookmark the school spreadsheet URL for quick access during the school day.

Reading the Attendance Record

Each row in the Attendance sheet represents one staff member's record for one day:

Field	Description
Date	Date of the record, in YYYY-MM-DD format.
Staff Name	The staff member's full name.
Staff ID	The staff member's unique ID.
Time-In	Time of the first scan of the day (HH:MM format).
Time-Out	Time of the last scan of the day (HH:MM format). Blank if not yet scanned out.
Status	Computed automatically. See Section 5.4 below.
Column 7	Used internally by the system for scan deduplication. Do not delete, edit, or hide it.

Understanding Status Values

The Status column is computed automatically every time a staff member scans, and is finalised by the nightly job after the school day ends.

Status	Meaning
On-Time	Staff member arrived and left within the allowed grace window.

Status	Meaning
Late	Time-In was after the allowed start grace period.
Left-Early	Time-Out was before the allowed end grace period.
Late Left-Early	Arrived late and also left early.
No Check-Out	Scanned in but did not scan out before the nightly job ran.
Late No Check-Out	Arrived late and also did not scan out.
Absent	No scan was recorded for the entire day. Inserted automatically by the nightly job.

Ongoing Admin Tasks

Adding a New Staff Member

Follow the same process as Chapter 4. In brief:

- Open the school spreadsheet and go to the Enrollment tab.
- Add a new row with the staff member's ID, Name, Role (**staff**), and Command (**register**).
- Have the staff member stand at the device and wait for the yellow LED.
- Guide them through the two finger placements.
- Confirm the Status reads **registered** before the staff member leaves.

Removing a Staff Member

When a staff member leaves the school, remove their fingerprint from the device so it can no longer be used.

- Open the Enrollment tab of the school spreadsheet.
- Find the staff member's row and note their FingerID number.
- In that row, change the Command cell to **delete** and clear the Status cell.
- The device will pick up the command within 10 seconds and delete the fingerprint. The Status column will update to **deleted**.

◆ NOTE

Deleting a fingerprint does not remove the staff member's historical attendance records. All past records remain in the Attendance sheet.

■ WARNING

Once deleted, the staff member cannot scan in again until re-enrolled. Only delete staff who have genuinely left.

Replacing a Lost or Changed Fingerprint

If a staff member can no longer be recognised — due to a cut, injury, or switching to a different finger — delete their current fingerprint and register a new one.

- Follow Section 6.2 to delete the existing fingerprint. Wait for the Status to update to **deleted**.
- In the same row, change the Command back to **register**, clear the Status cell, and clear the FingerID cell.
- Have the staff member stand at the device and follow the enrollment steps in Chapter 4 from Step 4 onwards.

Changing the School WiFi Network

If your school WiFi name or password changes, the device will lose its connection and the LED will glow steady purple. You will need the master finger to reconfigure it.

1. Make sure the device is powered on.
2. Place the master finger on the sensor. The LED will flash yellow five times, then turn purple — the WiFi setup portal is now running.
3. On your phone or laptop, connect to the **Attendance-Setup** WiFi network (password: setup1234).
4. Open a browser and go to **http://192.168.4.1**.
5. Enter the new WiFi name and password, then click **Save & Connect**.
6. The device will restart and the LED should return to steady blue.

Reviewing the Monthly Email Report

On the 1st of each month, an email is automatically sent to all configured admin addresses with the attendance summary for the previous calendar month.

The email contains:

- Staff members with absences, and how many days they were absent.
- Staff members with late arrivals, and how many times.
- Staff members with early departures.
- Staff members with missing check-outs.

The email also includes an Excel (.xlsx) attachment containing the full raw attendance log for the previous month and a Monthly Summary tab with aggregated counts per staff member — useful for payroll or HR reviews.

◆ NOTE

When the monthly report runs, the previous month's attendance rows are archived to a separate tab inside the spreadsheet (labelled "Archive - Month Year") and then deleted from the main Attendance sheet to keep it lean and fast. The data is preserved both in the archive tab and in the Excel attachment that was emailed to you. If you need to look up a past record, check either the archive tab in the spreadsheet or the relevant month's email attachment.

✓ TIP

If you do not receive the monthly email, check your spam or junk folder first. If it is not there, contact your device manufacturer.

Correcting a Wrong Record

If a record needs to be corrected — for example a Time-In value is wrong due to a clock error, or a staff member's name was misspelled — you can edit the Attendance sheet directly.

- Click the cell you need to change and type the correct value.
- For Time-In and Time-Out, use the format **HH:MM** (e.g. 07:28, 15:45).
- For Date, use the format **YYYY-MM-DD** (e.g. 2026-04-08).

- The system does not re-validate manually edited cells — whatever you type is saved as-is.

Checking for Missed Check-Outs

The nightly job automatically fills in **No Check-Out** for any staff member who scanned in but never scanned out. If you want to check earlier in the day, open the Attendance sheet and look for rows where the Time-Out column is blank and the Date is today — those staff members have arrived but not yet scanned out.

◆ NOTE

Offline scans are queued on the device and sent when WiFi reconnects. If a staff member scanned while the device was offline and a queued scan arrives after an Absent row was already written by the nightly job, the system will automatically correct the Absent row to a proper Time-In record. No manual action is needed.

Troubleshooting

Device LED Colour Guide

LED State	Meaning	Action
Steady blue	Connected to WiFi, ready.	None — normal operation.
Steady purple	No WiFi connection.	Reconfigure WiFi using master finger (Section 6.4).
Yellow flash ×5	Master finger detected — entering WiFi setup.	Proceed with setup if intended; otherwise unplug and replug to restart.
Solid yellow	Waiting for fingerprint during enrollment.	Place the correct finger on the sensor.
Green flash	Scan accepted / enrollment succeeded.	None.
Red flash	Finger not recognised, or enrollment failed.	Try again. If persistent, re-enrol (Section 6.3).

Common Problems and Solutions

Scans are not appearing in the Attendance sheet

- Check the device LED is steady blue. If it is purple, the device has no WiFi — reconfigure using the master finger (Section 6.4).
- The device queues scans offline and sends them once WiFi is restored. After reconnecting, wait a minute and check if queued scans appear.
- If the LED is blue but records still do not appear, contact your device manufacturer with the school name and the time the scan was made.

A staff member's finger is not being recognised

- Ask the staff member to wipe the sensor surface with a clean dry cloth, then try again.
- Ensure the staff member places the centre of their fingerprint on the sensor — not the tip of the finger.
- If the problem continues after several attempts, delete and re-enrol the fingerprint (Section 6.3).

Enrollment is stuck — the Status column never updates

- The device waits up to 20 seconds for a finger during enrollment. If no finger was placed in time it times out. To retry: clear the Status cell and change the Command back to register.
- Confirm the device LED is steady blue (online) before retrying.
- Check the Note column — the device may have written an error code there.

The device is not picking up the enrollment command

- The device polls for new commands every 10 seconds. Wait at least 15 seconds after adding the row.
- Confirm the Command cell contains exactly register, delete, or clearall — no capital letters, no extra spaces.
- Confirm the Status cell is blank or contains pending. If it says processing, the device has already picked it up — just wait.

The master finger is triggering the WiFi portal unexpectedly

- This is normal behaviour — placing the master finger always launches the portal. To cancel, unplug and replug the device to restart it.
- If this keeps happening accidentally, consider enrolling the master finger using a less-used finger (e.g. little finger) to avoid accidental triggers.

A staff member was marked Absent but they did scan in (offline scan)

- This can happen if the device was offline when the staff member scanned, and the nightly job ran before WiFi was restored.
- The system automatically corrects this: when the queued scan finally uploads, it will overwrite the Absent row with the correct Time-In. No manual action is needed.
- If the correction does not appear within a few minutes of WiFi reconnecting, contact your device manufacturer.

The monthly email was not received

- Check your spam or junk folder.
- Confirm your email address was provided to your device manufacturer to be added to the recipient list.
- Note: the email is sent on the 1st of each month.
- If neither helps, contact your device manufacturer.

◆ NOTE

If a problem is not covered in this guide, contact your device manufacturer and provide: the school name, a description of the issue, the LED colour or state at the time, and — if relevant — the exact text shown in the Status and Note columns of the Enrollment sheet.

Device Manufacturer Contact Details

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